

Covariance, Linear Regression & Regularization

IOAI Preparation — Advanced Track

IOAI Preparation

2026

*This problem set is aimed at students preparing for AI olympiads.
Theoretical questions are guided; the final problem requires genuine independent thinking.*

Overview. This problem set covers three fundamental ideas of machine learning:

1. **Covariance and geometry** — measuring how two quantities vary together, and what this means in vector space.
 2. **Linear regression** — finding the best-fit line through a cloud of points, and building it from data.
 3. **Regularization (Ridge & Lasso)** — preventing a model from memorizing noise.
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1 Covariance, Correlation, and Geometry

Definition — Mean and Variance (reminder)

Let $X = (x_1, \dots, x_n)$ be a data series.

$$\mathbb{E}[X] = \bar{X} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2.$$

Definition — Covariance

For two data series X and Y of length n :

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y}).$$

Intuition

Think of X = height (cm) and Y = weight (kg) of n people. When someone is tall ($x_i - \bar{X} > 0$) **and** heavy ($y_i - \bar{Y} > 0$), their contribution to the sum is positive. When tall people tend to be heavy and short people tend to be light, most terms are positive and $\text{Cov}(X, Y) > 0$.

Questions

1. **Sign of covariance.** For each pair, guess the sign of $\text{Cov}(X, Y)$ and justify intuitively.

- (a) X = study hours, Y = exam grade.
- (b) X = outside temperature, Y = heating consumption.
- (c) X = shoe size, Y = math grade.

2. **Basic properties.**

- (a) Show that $\text{Cov}(X, X) = \text{Var}(X)$.
- (b) Show that $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$, where $\mathbb{E}[XY] = \frac{1}{n} \sum x_i y_i$.
Hint: Expand $(x_i - \bar{X})(y_i - \bar{Y})$.
- (c) Deduce that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

3. **Numerical computation.**

i	1	2	3	4	5
x_i (study hours)	1	2	3	4	5
y_i (grade /20)	8	10	11	14	17

- (a) Compute $\bar{X}$, $\bar{Y}$, $\text{Var}(X)$, $\text{Var}(Y)$, and $\text{Cov}(X, Y)$.
- (b) Comment on the sign obtained.

4. **Pearson correlation coefficient.**

Covariance depends on units. We normalize it:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}.$$

- (a) Compute r for the data above.
- (b) What value of r corresponds to perfect correlation? To no linear relationship?
- (c) Comment: “ $r = 0.9$ between ice cream sales and drownings, so ice cream causes drowning.”

1.1 Geometric Interpretation: Correlation as a Cosine

Definition — Centered vectors

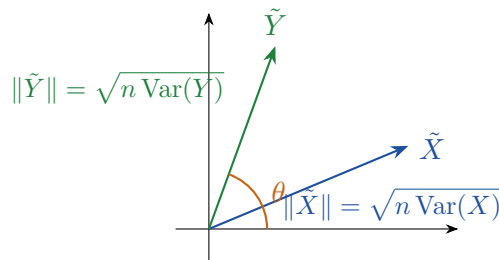
Given a data series $X = (x_1, \dots, x_n)$, define its **centered version**:

$$\tilde{X} = (x_1 - \bar{X}, x_2 - \bar{X}, \dots, x_n - \bar{X}) \in \mathbb{R}^n.$$

Note that $\|\tilde{X}\|^2 = \sum_{i=1}^n (x_i - \bar{X})^2 = n \text{Var}(X)$.

Intuition

Each data series of length n becomes a **vector in $\mathbb{R}^n$** once we subtract its mean. The angle θ between two such vectors encodes how aligned they are — and this is exactly the correlation.



5. Correlation as a cosine.

Recall that for two vectors $u, v \in \mathbb{R}^n$, the dot product satisfies $\langle u, v \rangle = \|u\| \|v\| \cos \theta$, where θ is the angle between them.

- Express $\langle \tilde{X}, \tilde{Y} \rangle$ in terms of n , $\text{Cov}(X, Y)$.
- Express $\|\tilde{X}\|$ and $\|\tilde{Y}\|$ in terms of n , $\text{Var}(X)$, $\text{Var}(Y)$.
- Deduce that:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} = \cos \theta,$$

where θ is the angle between $\tilde{X}$ and $\tilde{Y}$ in $\mathbb{R}^n$.

- Conclude that $-1 \leq r \leq 1$. When does $r = 1$? When does $r = 0$?
- What is the geometric meaning of $\text{Cov}(X, Y) = 0$?

6. Is transitivity of positive correlation true?

Suppose $\text{Cov}(X, Y) > 0$ and $\text{Cov}(Y, Z) > 0$. Must we have $\text{Cov}(X, Z) > 0$?

- Reformulate this question in terms of angles between $\tilde{X}$, $\tilde{Y}$, $\tilde{Z}$ in $\mathbb{R}^n$.
- Draw a picture in 2D with three vectors satisfying the angle conditions. What does it suggest?
- Find a concrete counterexample.** Construct three data series X , Y , Z of length n (you choose n) such that $\text{Cov}(X, Y) > 0$, $\text{Cov}(Y, Z) > 0$, but $\text{Cov}(X, Z) < 0$.
Hint: Think geometrically first. Can you place three vectors in $\mathbb{R}^2$ such that $\theta(\tilde{X}, \tilde{Y}) < 90^\circ$, $\theta(\tilde{Y}, \tilde{Z}) < 90^\circ$, but $\theta(\tilde{X}, \tilde{Z}) > 90^\circ$? What is the maximum possible angle between $\tilde{X}$ and $\tilde{Z}$?
- What can you say if additionally $\text{Cov}(X, Z) > 0$ is *required* by some extra condition? Try to give a sufficient condition on the angles.

Olympiad Challenge ★

Cauchy-Schwarz and the bound $|r| \leq 1$.

Using the Cauchy-Schwarz inequality on $\tilde{X}$ and $\tilde{Y}$, give a purely algebraic proof that $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}$, and characterize equality.

2 Linear Regression

Intuition

We have a cloud of points (x_i, y_i) and want to find the line $y = ax + b$ that fits them best. “Best” means the prediction errors are as small as possible.

Definition — Least-Squares Loss

$$\mathcal{L}(a, b) = \frac{1}{n} \sum_{i=1}^n (ax_i + b - y_i)^2.$$

Questions

1. Why squared errors?

- (a) Why not minimize $\frac{1}{n} \sum (ax_i + b - y_i)$ (without squaring)?
- (b) Why is $\frac{1}{n} \sum |ax_i + b - y_i|$ mathematically less convenient?

2. Computing the optimal coefficients.

- (a) Compute $\frac{\partial \mathcal{L}}{\partial b}$ and show $\hat{b} = \bar{Y} - \hat{a}\bar{X}$.
- (b) Compute $\frac{\partial \mathcal{L}}{\partial a}$ and show:

$$\hat{a} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Hint: Substitute $\hat{b}$ into the expression.

- (c) Covariance appears naturally. Explain qualitatively why this makes sense.

3. Numerical application.

Using the data from Section 1, compute $\hat{a}$ and $\hat{b}$, then:

- (a) Predict the grade for a student who studied 6 hours.
- (b) What happens if you extrapolate to 100 hours? Comment.

4. Multivariate regression.

With d features $x^{(1)}, \dots, x^{(d)}$, the model is $f_{\theta}(x) = \theta_0 + \sum_j \theta_j x^{(j)}$. In matrix form, let

$\tilde{x}_i = (1, x_i^{(1)}, \dots, x_i^{(d)})^\top$ and $X \in \mathbb{R}^{n \times (d+1)}$ be the design matrix.

- (a) Rewrite $\mathcal{L}(\theta) = \frac{1}{n} \|X\theta - Y\|^2$.
- (b) The analytical solution is $\hat{\theta} = (X^\top X)^{-1} X^\top Y$. When does this formula break down?

Olympiad Challenge ★

Unbiasedness of $\hat{a}$.

Suppose the true model is $Y_i = ax_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d. Show that $\mathbb{E}[\hat{a}] = a$, i.e., $\hat{a}$ is an unbiased estimator of a .

Hint: Write $\hat{a}$ as a function of a and the ε_i , then take the expectation.

3 Hands-on: Regression on Real Data

Hands-on

Dataset: [Advertising.csv](https://trevorhastie.github.io/ISLR/Advertising.csv) — direct download from the ISLR book authors (Trevor Hastie & Gareth James).

```
import pandas as pd
df = pd.read_csv("https://trevorhastie.github.io/ISLR/Advertising.csv",
index_col=0)
```

The dataset contains 200 products with four columns:

- **TV:** advertising budget spent on TV (\$k)
- **radio:** advertising budget on radio (\$k)
- **newspaper:** advertising budget in newspapers (\$k)
- **sales:** number of units sold (thousands)

The goal is to predict **sales** from the advertising budgets.

1. Exploration.

- (a) Load the data and compute the Pearson correlation between each of **TV**, **Radio**, **Newspaper** and **Sales**. Which feature is the most predictive?
- (b) Fit a simple linear regression of **Sales** on **TV** alone. Report $\hat{a}$, $\hat{b}$, and the MSE on the full dataset. Interpret $\hat{a}$: what does a \$1k increase in TV budget mean for sales?
- (c) Plot the scatter **TV** vs **Sales** and draw the regression line.

2. Multivariate regression.

- (a) Now use all three features (**TV**, **Radio**, **Newspaper**). Split 80/20 train/test. Report train MSE and test MSE.
- (b) Compare to using only **TV**. Does adding features always improve test performance?
- (c) The coefficient on **Newspaper** is likely small or negative in the multivariate model, even though $r(\text{Newspaper}, \text{Sales}) > 0$. Explain this apparent contradiction using the concept of **confounding**: which other variable is correlated with **Newspaper**?

3. Feature engineering.

- (a) Create a new feature: **TV_Radio** = **TV** $\times$ **Radio** (interaction term).

- (b) Retrain the model with features `TV`, `Radio`, `Newspaper`, `TV_Radio`. Does test MSE improve?
- (c) Explain intuitively what an interaction term captures. Why might `TV` and `Radio` *together* be more effective than their individual effects summed?
- (d) Now create 20 random noise features $Z^{(1)}, \dots, Z^{(20)} \sim \mathcal{N}(0, 1)$ and add them to the model. What happens to train MSE? To test MSE? Explain.

4. Thinking about features.

- (a) Suppose you also have access to: `season` (Q1/Q2/Q3/Q4), `competitor_spend` (\$k), `product_age` (years). Propose two additional feature engineering ideas that might improve the model. Justify each.
- (b) Is TV^2 a valid feature in a *linear* regression? (Recall: “linear” refers to linearity in the *parameters* θ , not in the raw inputs.)

4 Measuring Error: RMSE and the Bias-Variance Tradeoff

Definition — MSE and RMSE

Given predictions $\hat{y}_i$ and true values y_i on a test set of size n :

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}.$$

RMSE is expressed in the **same unit as** y , making it directly interpretable. A model predicting house prices in \$ with $\text{RMSE} = 20,000$ is off by \$20k on average.

Intuition

Why RMSE and not just the average error $\frac{1}{n} \sum (\hat{y}_i - y_i)$?

Because positive and negative errors cancel. A model that is +\$50k off on half the houses and -\$50k off on the other half has average error = 0 — but it’s clearly terrible. Squaring forces all errors to count positively.

The Bias-Variance Tradeoff

Definition — Bias and Variance of an estimator

Suppose the true relationship is $Y = f(X) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$. For a model $\hat{f}$ trained on a dataset, the expected prediction error at a point x decomposes as:

$$\underbrace{\mathbb{E}[(\hat{f}(x) - Y)^2]}_{\text{Expected MSE at } x} = \underbrace{\left(\mathbb{E}[\hat{f}(x)] - f(x)\right)^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}}.$$

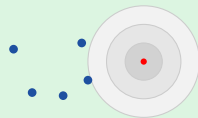
- **Bias**: how far off is the model *on average*? (systematic error)
- **Variance**: how much does the model change when trained on different datasets? (sensitivity to data)
- **Irreducible noise** σ^2 : the noise in the data itself — no model can beat this.

Intuition

Concrete analogy — archery.



Low bias, low variance



Low bias, high variance



High bias, low variance



High bias, high variance

Each dot is the model's prediction on a new dataset. The red center is the true value $f(x)$.

- **Bias** = how far the cluster is from the center.
- **Variance** = how spread out the cluster is.

The ideal is top-left: tight cluster centered on the truth.

Warning

The fundamental tension. Reducing bias usually increases variance and vice versa.

- A **simple model** (e.g., $f(x) = ax + b$) has *high bias* (can't fit complex patterns) but *low variance* (stable across datasets).
- A **complex model** (e.g., degree-15 polynomial) has *low bias* (fits anything) but *high variance* (wildly different on each new dataset).

You cannot make both simultaneously zero. The goal is to find the sweet spot.

Questions

1. Concrete examples.

- (a) A model always predicts the mean $\bar{Y}$ regardless of X . What are its bias and variance? Is its RMSE on new data always large?
- (b) A polynomial of degree $n - 1$ passes through all n training points exactly. What are its MSE on the training set and its expected bias? What about its variance?
- (c) You train a linear regression on 5 data points and obtain $\text{RMSE}_{\text{train}} = 0.3$, $\text{RMSE}_{\text{test}} = 4.1$. Diagnose the problem: high bias or high variance?

2. Effect of dataset size.

- (a) For a fixed model complexity, does increasing n (more training data) tend to reduce bias, variance, or both? Justify intuitively.
- (b) For a fixed n , does increasing model complexity (e.g., adding polynomial features) tend to reduce bias, variance, or both?
- (c) Sketch the curve of $\text{RMSE}_{\text{train}}$ and $\text{RMSE}_{\text{test}}$ as a function of model complexity (from very simple to very complex). Identify the sweet spot.

3. Where does regularization fit in?

Recall that Ridge shrinks the coefficients toward zero, effectively reducing model complexity.

- (a) Does Ridge increase bias, decrease bias, or leave it unchanged compared to ordinary least squares?
- (b) Does Ridge increase variance, decrease variance, or leave it unchanged?
- (c) Explain in one sentence why Ridge can reduce test RMSE even though it increases training MSE.

5 Overfitting and Regularization

Intuition

A model that *overfits* memorizes the training data — including its noise — and fails on new data.

Analogy: A student who memorizes every past exam answer verbatim will ace the same exam but fail a slightly different one.

Definition — Regularization

$$\mathcal{L}_{\text{reg}}(\theta) = \underbrace{\frac{1}{n} \sum_{i=1}^n (f_{\theta}(x_i) - y_i)^2}_{\text{fit}} + \lambda \cdot \underbrace{R(\theta)}_{\text{penalty}},$$

where $\lambda \geq 0$ controls regularization strength.

- **Ridge** (L^2): $R(\theta) = \|\theta\|_2^2 = \sum_j \theta_j^2$.
- **Lasso** (L^1): $R(\theta) = \|\theta\|_1 = \sum_j |\theta_j|$.

Questions

1. Role of λ .

- (a) What happens when $\lambda = 0$? When $\lambda \rightarrow +\infty$?
- (b) How would you choose λ in practice?

2. Ridge Regression.

In dimension 1, the Ridge loss is:

$$\mathcal{L}_{\text{Ridge}}(a) = \frac{1}{n} \sum_{i=1}^n (ax_i + \hat{b} - y_i)^2 + \lambda a^2.$$

- (a) Compute $\frac{\partial \mathcal{L}_{\text{Ridge}}}{\partial a}$ and find $\hat{a}_{\text{Ridge}}$.
- (b) Show that:

$$\hat{a}_{\text{Ridge}} = \frac{\text{Cov}(X, Y)}{\text{Var}(X) + \lambda}.$$

- (c) Compare to $\hat{a}$. What does regularization do to the coefficient?
- (d) **In dimension d :** the Ridge solution is $\hat{\theta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top Y$. Why does this fix the invertibility problem from Question 2.4b? (**Hint:** Think about the eigenvalues of $X^\top X + \lambda I$.)

3. Lasso and sparsity.

The Lasso penalizes by $\|\theta\|_1$, which has the remarkable property of setting some coefficients to *exactly* zero.

- (a) Give a concrete scenario (e.g., in genomics or text processing) where having exactly zero coefficients for many features is highly useful.
- (b) In what sense does Lasso perform implicit **variable selection**?
- (c) Compare: Ridge reduces all coefficients smoothly but never exactly to zero; Lasso can eliminate features entirely. When is each preferable?

4. Geometric comparison: Ridge vs Lasso.

Intuition

Regularization can be seen as a **geometric constraint**: minimize the error subject to staying inside a region.

- Ridge: $\theta_1^2 + \theta_2^2 \leq C$ — a **disk**.
- Lasso: $|\theta_1| + |\theta_2| \leq C$ — a **diamond**.

The diamond has corners on the axes ($\theta_1 = 0$ or $\theta_2 = 0$). The solution tends to “land” on a corner.

- (a) Draw both constraint regions in 2D for $C = 1$.
- (b) Explain geometrically why Lasso zeros out coefficients but Ridge does not.

5. **Numerical application.** Using Section 1 data, compute $\hat{a}_{\text{Ridge}}$ for $\lambda \in \{0, 0.5, 2, 10\}$ and comment on the evolution.

Olympiad Challenge ★

Ridge as a Bayesian estimator.

Assume a prior $\theta \sim \mathcal{N}(0, \tau^2)$ on the slope and Gaussian noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

1. Write the posterior $p(\theta | X, Y) \propto p(Y | X, \theta) p(\theta)$.
2. Show that the MAP (maximum a posteriori) estimator $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \log p(\theta | X, Y)$ is exactly the Ridge estimator with $\lambda = \sigma^2 / \tau^2$.

6 Final Problem: Predicting House Prices

Hands-on

Dataset: *House Prices: Advanced Regression Techniques* on Kaggle (free account required). Download `train.csv` and use it as your dataset — split it yourself into train/test.

```
df = pd.read_csv("train.csv")
```

You are given data on houses with the following raw features (among 79 total — focus on these):

Feature	Description
<code>GrLivArea</code>	Above-ground living area (sq ft)
<code>YearBuilt</code>	Year the house was built
<code>OverallQual</code>	Overall material quality (1–10)
<code>TotalBsmtSF</code>	Total basement area (sq ft)
<code>GarageArea</code>	Garage area (sq ft)
<code>Neighborhood</code>	Categorical: 25 neighborhoods
<code>SalePrice</code>	Target: sale price (\$)

Split `train.csv` into 80% train / 20% test using `train_test_split`.

1. Exploratory data analysis.

- (a) Compute the correlation matrix between all numerical features and `SalePrice`. Which raw feature is most predictive?
- (b) Plot the distribution of `SalePrice`. Is it symmetric? What transformation might make it more suitable for linear regression, and why?
- (c) **Apply the transformation** $y \leftarrow \log(\text{SalePrice})$ and recompute correlations with the log-price. Do the correlations improve?

2. Baseline model.

- (a) Fit a linear regression using only `GrLivArea` to predict $\log(\text{SalePrice})$. Report train and test RMSE (in log-price units).

- (b) Add all numerical features. Does test RMSE improve? How does the gap between train and test RMSE change?

3. Feature engineering.

Think carefully about what drives house prices beyond the raw numbers.

- (a) Create the feature `HouseAge = 2025 - YearBuilt`. Does this improve over using `YearBuilt` directly? Why might it be more informative?
- (b) Create `TotalSF = GrLivArea + TotalBsmtSF`. What does this feature capture that the two separate features do not?
- (c) Create `QualArea = OverallQual × GrLivArea` (interaction term). Justify why this interaction makes economic sense.
- (d) Encode the `Neighborhood` categorical variable using **one-hot encoding** (25 dummy variables). How many parameters does this add to the model? Is this a concern?
- (e) Retrain the model with all engineered features. Report train and test RMSE. Is there improvement?

4. Regularization.

- (a) With all engineered features (including the 24 neighborhood dummies), apply Ridge regression with $\lambda \in \{0.01, 0.1, 1, 10, 100\}$. For each λ , report train and test RMSE. Draw the regularization path.
- (b) Do the same with Lasso. How many features does Lasso eliminate for $\lambda = 1$? Are the eliminated features interpretable (noise vs. signal)?
- (c) Use 5-fold cross-validation to select the best λ for Ridge. Report the chosen value and the corresponding test RMSE.
- (d) Compare the coefficients of the neighborhood dummies under Ridge vs Lasso. What happens qualitatively?

5. Critical thinking.

- (a) Your model achieves a test RMSE of 0.12 in log-price. Convert this to a rough dollar error: if the median price is \$180,000, what is the approximate error in dollars? (**Hint:** $e^{0.12} - 1 \approx 0.127$.)
- (b) You observe that your model systematically underpredicts very expensive houses. What might cause this, and what could you do about it?
- (c) Suppose you add a feature `is_renovated` (binary: was the house renovated after construction?). This feature has high correlation with price in the training set, but you suspect it is not available at prediction time for some houses. Discuss the trade-off.
- (d) (**Bonus**) A friend suggests adding `SaleCondition` (normal/abnormal/partial sale) as a feature, arguing it is highly correlated with price. Why might this be a form of **data leakage**, and why is it dangerous?

Remark

Going further.

- Ridge corresponds to a Gaussian prior on θ (shown in the challenge above); Lasso corresponds to a Laplace prior.
- **Elastic Net** combines both: $\lambda_1 \|\theta\|_1 + \lambda_2 \|\theta\|_2^2$.
- In high dimension ($d \gg n$), regularization is not optional — it is a *necessity* for any meaningful generalization.
- The bias-variance tradeoff formalizes exactly what we observed: more complex models reduce bias but increase variance.